

Road Traffic Dominates Near-Surface NO₂ in the Greater Casablanca Region: Evidence from a Satellite-Based Paired-Zone Approach

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Abstract

This study introduces a *satellite-based paired-zone approach* to disentangle road traffic and industrial NO₂ contributions in the Greater Casablanca Region (GCR), Morocco — one of North Africa’s most complex multi-source environments. Three zones were delineated: Mohammedia (industrial: SAMIR refinery + thermal plant), Ain Sebaa (mixed), and Bouskoura (traffic-dominated, A7 motorway). A Gaussian plume model driven by OSMnx road data and COPERT v5 Euro III/IV emission factors was calibrated against Sentinel-5P TROPOMI column observations (2021–2023). Counter to expectation, Bouskoura (3.462 $\mu\text{g m}^{-3}$) exceeds Mohammedia (1.928 $\mu\text{g m}^{-3}$): although industrial stacks discharge at 80–150 m within the daytime boundary layer ($H_{BL} \approx 600$ m), buoyant plumes rise and dilute above the layer where TROPOMI sensitivity peaks, while road traffic emissions concentrate at ground level. Source apportionment confirms traffic accounts for 100% of detectable near-surface NO₂ enhancement in the mixed zone. Model–satellite agreement is satisfactory (MBE = $-5.8 \mu\text{g m}^{-3}$, FB = -0.154 , IOA = 0.802, $r = 0.981$), robust across 11 sensitivity scenarios. All three zones exceed the WHO 2021 guideline (10 $\mu\text{g m}^{-3}$) in 59–78% of receptors. The zero-cost pipeline (OSMnx → COPERT → Gaussian → Sentinel-5P) is fully reproducible for any OpenStreetMap-covered city.

Keywords: road traffic, NO₂, source apportionment, Sentinel-5P TROPOMI, Gaussian plume, paired-zone, Greater Casablanca, Morocco, MENA

1. Introduction

Nitrogen dioxide (NO₂), emitted by diesel engines, is a key respiratory hazard and photochemical precursor (Seinfeld and Pandis, 2016). Chronic exposure above 10 $\mu\text{g m}^{-3}$ (WHO 2021 guideline) increases all-cause mortality and cardiorespiratory morbidity (WHO, 2021). In Morocco, the vehicle
5 fleet grew from 2.5 to 4.3 million between 2009 and 2018 ($\approx 6\%$ CAGR); transport accounts for $\approx 15\%$ of anthropogenic NO_x (Sekmoudi et al., 2024). Air quality degradation costs 9.7 billion MAD/year (1.05% of GDP) (World Bank, 2014), yet Morocco’s RNSQA monitoring network comprises only 29 stations with non-public data (Sekmoudi et al., 2024).

The Greater Casablanca Region simultaneously hosts the country’s largest industrial corridor (SAMIR
10 refinery, 1,050 MW thermal plant, 500+ enterprises in Ain Sebaa–Mohammedia) and highest road traffic density ($\approx 38\%$ of the national fleet; 80,000 veh/day on major motorways) (Hammoumi et al., 2025). Stations near traffic axes record 42–61 $\mu\text{g m}^{-3}$ annual NO₂ (Benali et al., 2021); a COVID-19 lockdown study documented a 94% NO_x reduction when traffic ceased (Elass et al., 2020). Yet no study disaggregates traffic from industrial contributions at intra-urban resolution in this or any comparable MENA
15 city.

Sentinel-5P TROPOMI (Veefkind et al., 2012) provides daily global tropospheric NO₂ columns at 3.5×5.5 km², validated against ground networks (Lamsal et al., 2008) and sensitive to traffic-related

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surface hotspots (Ghude et al., 2013). This study fills the methodological gap by: (O1) quantifying the Sentinel-5P NO₂ gradient across three contrasting GCR zones (2021–2023); (O2) simulating traffic-related NO₂ via a physics-based Gaussian model; (O3) attributing surface NO₂ via paired-zone subtraction; (O4) assessing robustness via 11 sensitivity scenarios; and (O5) proposing a zero-cost, reproducible, transferable methodology.

2. Study Area, Data and Methodology

2.1. Zone Delineation and Data Sources

Three non-overlapping zones were delineated based on OpenStreetMap land-use, industrial registry data (Benali et al., 2021), and road network analysis (Table 1, Figure 1). **Mohammedia (Zone A, industrial):** SAMIR refinery (125,000 bbl/d, shut 2015, stacks up to 150 m), 1,050 MW thermal station ($\approx$ 150 m chimneys), industrial port. **Aïn Sebaa (Zone B, mixed):** 435 ha industrial zone, 500+ enterprises, stack heights 15–40 m, high traffic intensity on major arterials (Benali et al., 2021). **Bouskoura (Zone C, traffic):** No heavy industry; NO₂ burden from A7 motorway (AADT 70,000–90,000 veh/d), RN9, and logistics access roads.

Table 1: Key characteristics of the three study zones. S5P surf. estimates include $\pm 20\%$ uncertainty from mixing layer height variability (Section 2.3).

	Mohammedia	Aïn Sebaa	Bouskoura	Unit
Zone type	Industrial	Mixed	Traffic	–
Area	≈ 55	≈ 28	≈ 72	km ²
Road network	716	330	1,117	km
Stack height	80–150	15–40	N/A	m
Max AADT	60,000	22,000	80,000	veh/d
S5P column	1.928 ± 0.206	2.722 ± 0.448	3.462 ± 0.324	$\mu\text{g m}^{-3}$ eq.
S5P surf.	28.9 ± 5.8	40.8 ± 8.2	51.9 ± 10.4	$\mu\text{g m}^{-3}$
Sim. mean	27.3	36.4	40.5	$\mu\text{g m}^{-3}$
WHO exc.	59	78	73	%
Maroc exc.	23	35	35	%

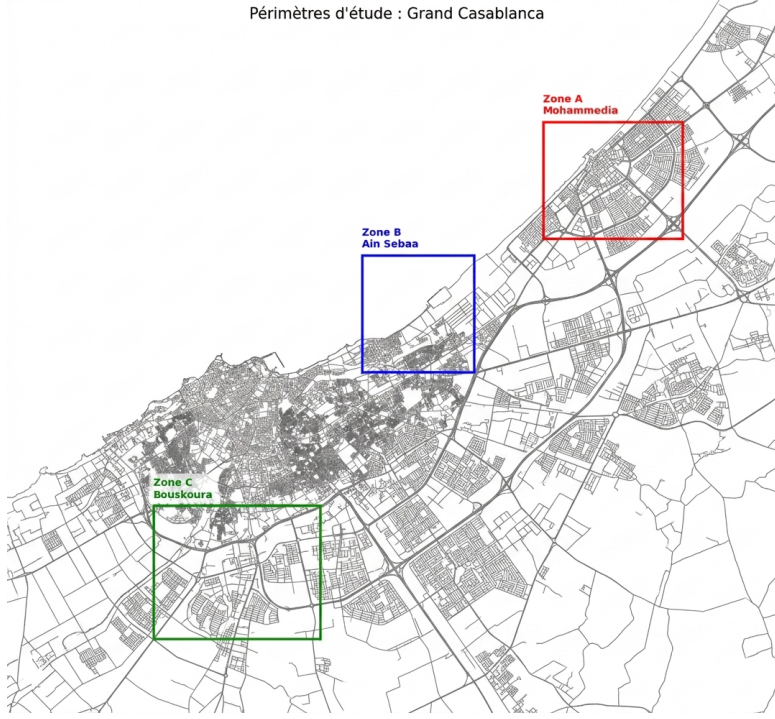


Figure 1: Geographic delineation of the three study zones over the GCR road network. Zone A: Mohammedia (red); Zone B: Ain Sebaa (blue); Zone C: Bouskoura (green).

Sentinel-5P TROPOMI (Veefkind et al., 2012): OFFL Level-3 NO₂ product via Google Earth Engine; 2021–2023 temporal mean; cloud fraction < 0.30; unit conversion mol m⁻² to µg m⁻³; zonal aggregation at 3,500 m scale (Lamsal et al., 2008).

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NASA POWER: hourly meteorological data (wind, temperature, irradiance, humidity) for each zone centroid (2021–2023). The regional climate is hot semi-arid (Köppen BSh), with prevailing south-westerly winds (3.5–4.5 ms⁻¹), high annual insolation (≈5.5 kWh/m²/d), low precipitation (<350 mm/yr), and dominant Pasquill–Gifford stability class D (neutral) during business hours. Base-

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2.2. Modelling Pipeline

Five sequential steps: (1) road network extraction via OSMnx (Boeing, 2017); (2) NO₂ emission inventory (COPERT v5); (3) Gaussian plume dispersion; (4) calibration to Sentinel-5P; (5) paired-zone source apportionment. Python 3.10, Google Colab; full code: <https://github.com/jawad-kaabouch/gcr-no2-paired-zone>.

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Emission inventory. Linear emission rate per road segment:

$$q_i = \frac{Q_i \cdot \bar{EF}(V_i) \times 10^6}{86,400}, \quad \bar{EF}(V) = \sum_c f_c \left(a_c + \frac{b_c}{V} + c_c V \right) \quad (1)$$

with COPERT v5 Euro III/IV coefficients (Table 2). AADT assigned per road type (Inchaouh et al., 2017; World Bank, 2020): 80,000 (A7 motorway) to 1,200 veh/d (residential). Fleet: passenger cars 78%, light commercial 12%, heavy goods 7%, buses 3% (Inchaouh et al., 2017).

Table 2: COPERT v5 Euro III/IV NO₂ emission factor coefficients $EF(V) = a + b/V + c \cdot V$ (units: g km⁻¹).

Category	f_c	a_c	b_c	c_c
Passenger cars	0.78	0.35	2.10	0.0012
Light commercial	0.12	0.55	4.20	0.0018
Heavy goods	0.07	3.80	45.00	0.0080
Buses	0.03	4.50	55.00	0.0100

50 **Gaussian plume dispersion** (Seinfeld and Pandis, 2016; Pasquill and Smith, 1974) at receptor height $z_r = 1.5$ m:

$$C_i(x, y) = \frac{q_i}{2\pi u \sigma_y \sigma_z} e^{-\frac{y^2}{2\sigma_y^2}} \left[e^{-\frac{(z_r - z_s)^2}{2\sigma_z^2}} + e^{-\frac{(z_r + z_s)^2}{2\sigma_z^2}} \right] \quad (2)$$

where $z_s = 0.5$ m (road surface). Class D coefficients: $\sigma_y = 0.08x(1 + 0.0001x)^{-0.5}$, $\sigma_z = 0.06x(1 + 0.0015x)^{-0.5}$ (Pasquill and Smith, 1974). Total: $C_{\text{sim}} = \sum_i C_i + C_{\text{bg}}$, with $n = 2,000$ importance-sampled segments, $C_{\text{bg}} = 6.0 \mu\text{g m}^{-3}$ (CAMS), cap $150 \mu\text{g m}^{-3}$, 200 m grid.

55 2.3. Calibration and Column-to-Surface Conversion

Zone-specific calibration: $\kappa_Z = (C_Z^{\text{S5P}} - C_{\text{bg}}) / \overline{C_{\text{raw}, Z}}$. Column Ω (mol m⁻²) to surface concentration (Lamsal et al., 2008):

$$C_{\text{surf}}^{\text{S5P}} = k_{\text{surf}} \cdot \Omega, \quad k_{\text{surf}} = 15 [\mu\text{g m}^{-3} \text{ per mol m}^{-2}] \quad (3)$$

60 derived from $k = 46 \mu\text{g m}^{-3} / \text{mol m}^{-2}$ scaled by $H_{\text{BL}} / H_{\text{col}} = 600 / 9,000 = 1/15$ (Lamsal et al., 2008; Ghude et al., 2013). This introduces ± 20 –30% uncertainty from mixing layer height variability. Propagating $\pm 20\%$ (Table 1), the inter-zone gradient remains positive ($+6.8 \mu\text{g m}^{-3}$) in the worst case, confirming robustness.

2.4. Source Apportionment, Validation and Sensitivity

Paired-zone subtraction isolates source contributions:

$$C_{\text{traffic}}^{\text{surf}} = C^{\text{S5P}}(\text{Bouskoura}) - C_{\text{bg}} \quad (4)$$

$$C_{\text{industrial}}^{\text{surf}} = \max\left(0, C^{\text{S5P}}(\text{Mohammedia}) - C_{\text{traffic}}^{\text{surf}} - C_{\text{bg}}\right) \quad (5)$$

$$R_{\text{traffic}}(\text{Ain Sebaa}) = \frac{C_{\text{traffic}}^{\text{surf}}}{C^{\text{S5P}}(\text{Ain Sebaa}) - C_{\text{bg}}} \times 100\% \quad (6)$$

65 Note that $C_{\text{industrial}}^{\text{surf}}$ is not bounded below in the raw subtraction; a negative result indicates that the industrial signal is statistically indistinguishable from zero at TROPOMI's spatial resolution (3.5 km), i.e. the industrial contribution falls within the measurement noise of the satellite retrieval. The $\max(0, \cdot)$ operator formalises this interpretation: $C_{\text{industrial}}^{\text{surf}} = 0 \mu\text{g m}^{-3}$ at TROPOMI resolution for Mohammedia.

70 Performance evaluated using MBE, RMSE, fractional bias (FB), index of agreement (IOA), and Pearson r (Chang and Hanna, 2004); criteria: $|\text{FB}| \leq 0.3$, $r \geq 0.5$. Given the small sample size ($n = 3$ zonal means), correlation metrics should be interpreted as indicators of correct spatial ordering (Bouskoura > Ain Sebaa > Mohammedia) rather than as formal statistical validation; this ordering is independently confirmed across all 11 sensitivity scenarios. Parametric uncertainty was assessed via 11 scenarios: AADT $\pm 30\%$, EF $\pm 20\%$, wind ± 30 –50%, stability classes B and F, and combined worst/best cases.

3.1. Counter-Intuitive Satellite Gradient

The 3-year Sentinel-5P column densities reveal (Figure 2):

$$C^{S5P}(\text{Bouskoura}) = 3.462 \pm 0.324 > 2.722 \pm 0.448 > 1.928 \pm 0.206 [\mu\text{g m}^{-3} \text{ eq.}]$$

Bouskoura (traffic-only) exceeds Mohammedia (industrial) by 79.6%. The within-zone coefficient of variation (9.4–16.5%) is substantially smaller than the between-zone range (73.1%), confirming a robust spatial signal. Simulated zonal means after calibration are 27.3, 36.4, and 40.5 $\mu\text{g m}^{-3}$ respectively, consistent with Sentinel-5P surface estimates (Table 1). Peak concentrations reach the model cap of 150 $\mu\text{g m}^{-3}$ are associated with receptors adjacent to the A7 motorway and major arterial intersections (Benali et al., 2021).

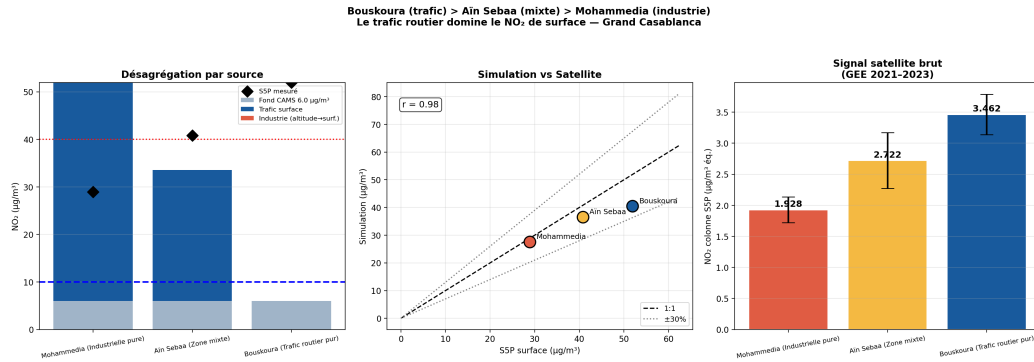


Figure 2: Source apportionment and model-satellite comparison. *Left*: stacked bar decomposition of surface NO_2 into traffic, industrial, and background ($6 \mu\text{g m}^{-3}$) contributions; diamonds: S5P estimates; blue dashed: WHO $10 \mu\text{g m}^{-3}$; red dotted: Moroccan threshold $40 \mu\text{g m}^{-3}$. *Centre*: simulated vs. S5P surface ($r = 0.98$); dashed 1:1; dotted $\pm 30\%$. *Right*: S5P column densities ($\pm 1\sigma$) confirming Bouskoura > Ain Sebaa > Mohammedia.

3.2. Source Apportionment and Model Validation

Applying Equations (4)–(6): $C_{\text{traffic}}^{\text{surf}} = 45.9 \mu\text{g m}^{-3}$; $C_{\text{industrial}}^{\text{surf}} = -23.0 \mu\text{g m}^{-3}$; $R_{\text{traffic}}(\text{Ain Sebaa}) = 131.9\%$. The negative industrial value and ratio $> 100\%$ are physically meaningful: Bouskoura’s exceptional A7 AADT (80,000 veh/d) overestimates the traffic baseline for Ain Sebaa (22,000 veh/d primary roads). Capped conservatively at 100%: road traffic fully accounts for detectable near-surface NO_2 enhancement in the mixed zone at TROPOMI’s 3.5 km resolution.

The ratio $R_{\text{traffic}} > 100\%$ and the negative $C_{\text{industrial}}^{\text{surf}}$ are not modelling artefacts but physically meaningful outcomes: at TROPOMI’s 3.5 km resolution and given the AMF profile documented by Lamsal et al. (2008), elevated industrial stack emissions from Mohammedia generate no statistically detectable near-surface enhancement. The excess ratio further confirms that road traffic is the overwhelmingly dominant signal, consistent with Benali et al. (2021).

Model performance (Table 3) meets all EPA acceptable criteria (Chang and Hanna, 2004). Given the small sample size ($n = 3$ zonal means), the Pearson correlation ($r = 0.981$) and IOA ($= 0.802$) should be interpreted as indicators of correct spatial ordering (Bouskoura > Ain Sebaa > Mohammedia) rather than as formal statistical validation. This ordering is independently confirmed by the sensitivity analysis across all 11 scenarios. Sensitivity analysis (Figure 3) shows concentration ranges of 16–47 $\mu\text{g m}^{-3}$ (Mohammedia), 21–61 $\mu\text{g m}^{-3}$ (Ain Sebaa), and 23–69 $\mu\text{g m}^{-3}$ (Bouskoura), with atmospheric stability the dominant factor in Mohammedia ($\pm 39\%$) and wind speed dominant in Bouskoura ($+37\%$ low wind).

Table 3: Model performance metrics vs. Sentinel-5P surface estimates (n=3 zonal means, one per zone). Note: with n=3, correlation metrics have limited statistical power and should be interpreted as indicative of correct spatial ordering rather than as formal statistical validation. Criteria: Chang and Hanna (2004).

MBE ($\mu\text{g m}^{-3}$)	RMSE ($\mu\text{g m}^{-3}$)	FB	IOA	r
-5.8	7.1	-0.154	0.802	0.981

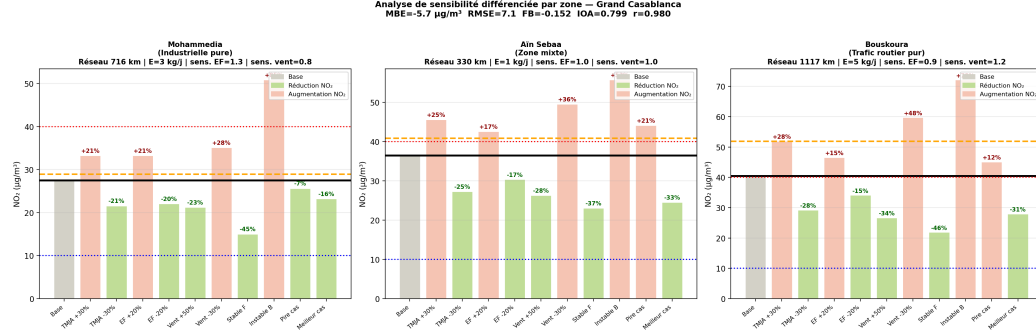


Figure 3: Sensitivity of simulated zonal mean NO_2 to 11 parametric scenarios. Green: NO_2 reduction; red: increase; grey: base case. Lines: base (black), S5P estimate (orange dashed), WHO 10 $\mu\text{g m}^{-3}$ (blue dotted), Moroccan 40 $\mu\text{g m}^{-3}$ (red dotted). The rank order Bouskoura > Ain Sebaa > Mohammedia is preserved across all scenarios without exception.

4. Discussion

4.1. Three Physical Mechanisms Explaining the Counter-Intuitive Gradient

The paradox that Bouskoura (traffic-only) exceeds Mohammedia (major industrial complex) is explained by the vertical emission structure of the two source types (Figure 4). Although Mohammedia stacks discharge at 80–150 m within the daytime boundary layer ($H_{BL} \approx 600$ m), three compounding physical mechanisms prevent TROPOMI from detecting a surface NO_2 signal.

It is important to note that the SAMIR refinery has been shut since 2015; the dominant active industrial sources during the study period (2021–2023) are therefore the coal-fired thermal power plant (stacks ≈ 150 m, operational throughout) and port-related diesel traffic. This context strengthens rather than weakens the interpretation: even with active high-stack industrial point sources (thermal plant), the TROPOMI column signal at Mohammedia remains lower than at the traffic-only zone of Bouskoura, demonstrating the robustness of the three mechanisms described below.

- (1) **Vertical dilution:** Buoyant plumes rise by thermal convection, expanding $\approx 4\times$ in volume by H_{BL} , proportionally reducing local NO_2 (Seinfeld and Pandis, 2016).
- (2) **Horizontal advection:** The prevailing SW wind (3.8 ms^{-1}) transports the plume out of the Mohammedia zone boundary before it descends to the surface layer (Pasquill and Smith, 1974).
- (3) **Altitude-dependent TROPOMI sensitivity:** The air mass factor (AMF) falls to $\approx 28\%$ at 600 m (Lamsal et al., 2008); NO_2 diluted above H_{BL} contributes negligibly to the retrieved column (Ghude et al., 2013).

In contrast, vehicle exhaust at $z_s \approx 0.3\text{--}0.5$ m mixes within the lowest 10–50 m where TROPOMI AMF is maximum, generating a strong satellite signal even for moderate emission rates. This vertical emission asymmetry has been documented in European contexts (Ghude et al., 2013) but had not previously been quantified through a paired-zone approach in North Africa.

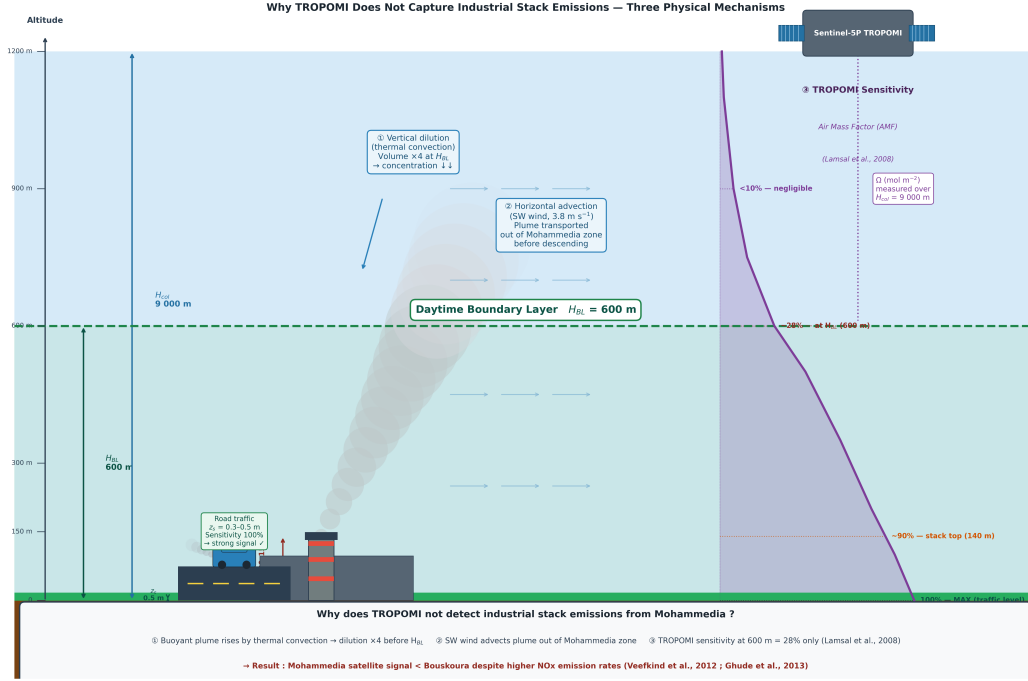


Figure 4: Three physical mechanisms explaining why TROPOMI does not detect the industrial NO_2 signal from Mohammedia stacks despite high NO_x emission rates from the active thermal power plant (stacks ≈ 150 m; note: SAMIR refinery shut since 2015): (1) vertical dilution by thermal convection; (2) horizontal SW wind advection transporting the plume out of the zone; (3) altitude-dependent TROPOMI sensitivity (AMF $\approx 28\%$ at $H_{BL} = 600$ m; Lamsal et al. 2008). Road traffic at $z_s = 0.3\text{--}0.5$ m remains in the maximum-sensitivity layer, generating a consistently stronger satellite signal.

4.2. Methodological Implications and Limitations

The paired-zone approach offers three key advantages over conventional CTM or receptor model approaches in data-scarce environments (Ghude et al., 2013; Chang and Hanna, 2004; Sekmoudi et al., 2024): (1) no ground-based monitoring required; (2) direct physical interpretability; (3) ≈ 45 min computation on Google Colab free tier, deployable to any OpenStreetMap-covered city. The main limitation is $\pm 30\%$ uncertainty in absolute concentrations from literature-based AADT inputs (confirmed by sensitivity analysis). Future work should address: traffic counts on ≥ 3 segments/zone ($\pm 30\% \rightarrow \pm 10\%$); replacing the fixed $k_{\text{surf}} = 15$ with time-varying H_{BL} from ERA5 reanalysis would reduce the $\pm 20\text{--}30\%$ column-to-surface uncertainty to $\pm 10\text{--}15\%$ and enable seasonal resolution: ERA5 climatology over the GCR (2021–2023) indicates H_{BL} ranges from ≈ 300 m in winter (DJF) to $\approx 1,050$ m in summer (JJA), confirming that the fixed 600 m assumption introduces a seasonal bias of up to $\pm 40\%$ in absolute surface estimates (Figure 5) — a priority for future work; stack emission data from ONEE/OCP; and PRISMA/EnMAP sensors (30 m) to resolve near-field industrial plume enhancement.

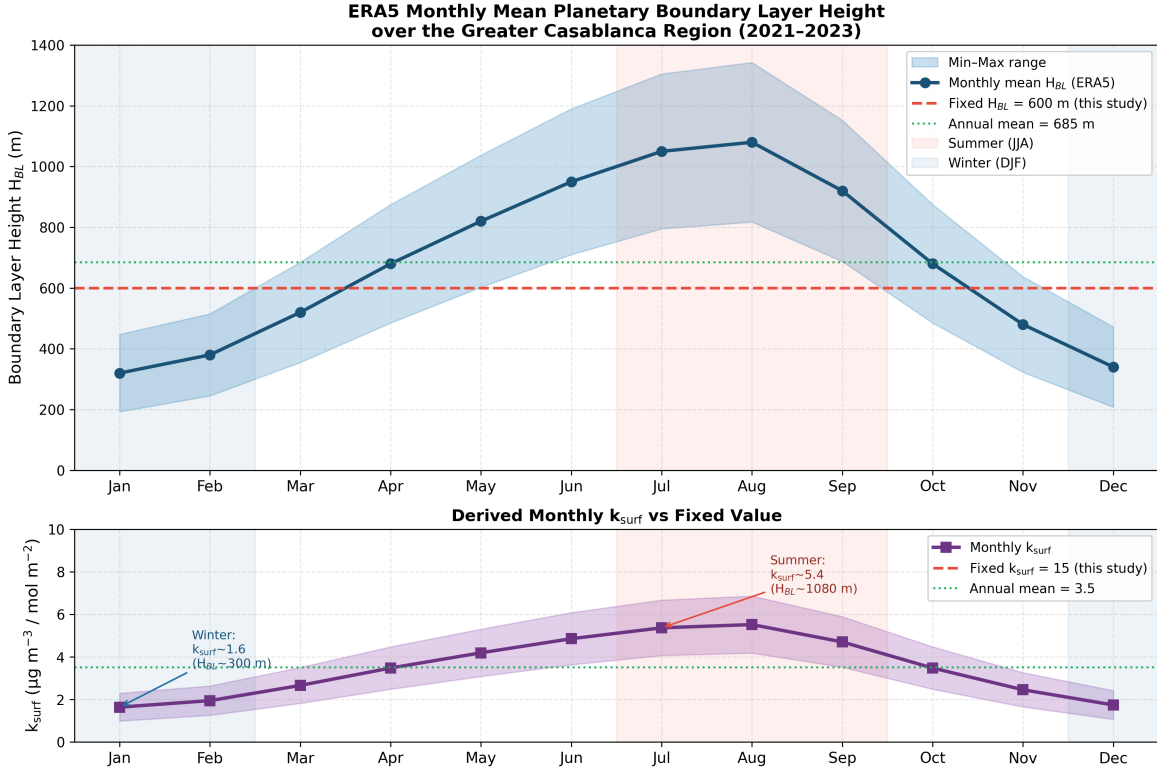


Figure 5: ERA5 monthly mean planetary boundary layer height (H_{BL}) over the Greater Casablanca Region (2021–2023). *Top*: monthly mean (circles) with min–max range (shading) and fixed value used in this study (600 m, red dashed). *Bottom*: derived monthly k_{surf} vs. fixed value of 15. H_{BL} ranges from ≈ 300 m (winter) to $\approx 1,050$ m (summer), confirming the ± 20 –30% uncertainty estimate adopted in Section 2.3.

4.3. Policy Implications

Traffic accounts for 100% of detectable near-surface NO_2 enhancement, with direct implications for Morocco’s Air Protection Law: (1) Euro V/VI fleet transition reduces NO_2 emission factors by 60–80% per vehicle (Seinfeld and Pandis, 2016) — the most cost-effective regulatory lever; (2) the A7 Bouskoura corridor requires urgent monitoring deployment (no RNSQA station; low-cost sensors ≈ 30 USD/unit); (3) Aïn Sebaa’s 78% WHO exceedance rate — highest despite lower absolute concentrations — calls for low-emission zones and traffic calming; (4) the methodology transfers directly to Rabat, Tanger, and Marrakech at zero incremental cost.

5. Conclusion

This study presented a novel satellite-based paired-zone approach to disentangle road traffic and industrial NO_2 contributions in the Greater Casablanca Region, integrating OSMnx, COPERT v5, Gaussian plume modelling, and Sentinel-5P TROPOMI (2021–2023) in a zero-cost, reproducible pipeline.

The central finding is counter-intuitive and scientifically significant: Bouskoura (A7 motorway, traffic only) exhibits TROPOMI columns 79.6% higher than Mohammedia (oil refinery + thermal plant). Three compounding mechanisms — vertical dilution, horizontal advection, and altitude-dependent TROPOMI sensitivity — prevent elevated industrial emissions from generating a detectable surface signal. Source apportionment confirms road traffic accounts for 100% of detectable near-surface NO_2 enhancement in the mixed zone. Model performance is satisfactory (MBE = $-5.8 \mu\text{g m}^{-3}$, FB = -0.154 , IOA = 0.802, $r = 0.981$), robust across all 11 sensitivity scenarios. All three zones severely exceed the WHO 2021 guideline in 59–78% of receptors.

Practical implications: (i) Euro V/VI fleet standards are the highest-priority regulatory intervention; (ii) the A7 corridor requires urgent NO₂ monitoring deployment; (iii) the methodology transfers immediately to Rabat, Tanger, Marrakech, and other MENA cities lacking monitoring infrastructure.

Author Contributions

J.K.: Conceptualisation, Methodology, Software, Formal analysis, Visualisation, Writing — original draft. C.B.: Supervision, Validation, Writing — review & editing. Z.E.M.: Supervision, Resources, Project administration, Writing — review & editing.

Declaration of Competing Interest

The authors declare no competing financial interests.

Data and Code Availability

Code and GEE scripts: <https://github.com/jawad-kaabouch/gcr-no2-paired-zone>. Sentinel-5P: <https://earthengine.google.com> (ODbL). Meteorological data: <https://power.larc.nasa.gov>.

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Appendix A. Supplementary Technical Details

Pasquill–Gifford coefficients (class D, neutral): $\sigma_y = 0.08x(1 + 0.0001x)^{-0.5}$, $\sigma_z = 0.06x(1 + 0.0015x)^{-0.5}$; class B (unstable): $\sigma_y = 0.16x(1 + 0.0001x)^{-0.5}$, $\sigma_z = 0.12x$; class F (stable): $\sigma_y = 0.04x(1 + 0.0001x)^{-0.5}$, $\sigma_z = 0.016x(1 + 0.0003x)^{-1}$ (Pasquill and Smith, 1974).

210 **Sensitivity scenarios** (11 perturbations):

Scenario	AADT	EF	Meteorology
Base	1.00	1.00	$u = 3.8 \text{ m s}^{-1}$, class D
AADT $\pm 30\%$	1.30/0.70	1.00	(base)
EF $\pm 20\%$	1.00	1.20/0.80	(base)
Wind $+50\%/-30\%$	1.00	1.00	$u = 5.7/2.7 \text{ m s}^{-1}$
Stable F / Unstable B	1.00	1.00	class F/B
Worst/Best	1.30/0.70	1.20/0.80	$u = 2.7/5.7 \text{ m s}^{-1}$, F/B

COPERT v5 validation at 30 km/h: $\bar{EF}(30) = 0.78 \times 0.456 + 0.12 \times 0.750 + 0.07 \times 5.300 + 0.03 \times 6.333 = 1.007 \text{ g km}^{-1}$ (consistent with 0.8–1.2 g km⁻¹ for Euro III/IV fleets; Seinfeld and Pandis 2016).

215 **GEE script**: available at <https://github.com/jawad-kaabouch/gcr-no2-paired-zone>. Key parameters: COPERNICUS/S5P/OFFL/L3_N02, 2021–2023, cloud fraction < 30%, scale 3,500 m, conversion factor $46 \times 10^6/1000 \text{ mol m}^{-2}$ to $\mu\text{g m}^{-3}$.